

Golden Calibration Log Analysis

We've plotted the data from `cal_6_calibration_2026-09-11_08-34-00-240.csv` to visualize the quality of the calibration and observe the sensor readings over the test duration.

Magnetometer Calibration (Hard and Soft Iron)

This visualization shows how the raw magnetometer readings (which form a distorted, off-center ellipsoid due to hard/soft iron interference) are transformed into a clean sphere centered on the origin.

```
![Raw Magnetometer Data](images/calibration/mag_raw_3d.png)
<!-- slide -->
![Calibrated Magnetometer Data](images/calibration/mag_cal_3d.png)
```

Observations:

- **Raw Data:** The reference and tip sensors have massive hard-iron offsets (the ellipsoids are very far from 0,0,0, particularly the reference sensor which is offset deeply in the -Z and +X directions, and the tip sensor offset deeply in the +X, +Y directions).
- **Calibrated Data:** The spheres are perfectly centered on the origin and are uniform in radius. This confirms the Python Kabsch alignment and soft/hard iron correction matrices are performing beautifully. The baseline magnetic field magnitude is consistently normalized to roughly 50,000 nT.

Coverage Assessment

The Python calibration tool evaluates how completely the sensor was rotated through all 3D axes. A higher coverage percentage ensures the resulting calibration sphere is accurate in all orientations.

```
--- Reference Sensor Coverage Assessment ---
Ideal Magnetic Radius: 50547 nT
X-Axis Coverage: 96.6%
Y-Axis Coverage: 99.7%
Z-Axis Coverage: 72.9%

--- Tip Sensor Coverage Assessment ---
Ideal Magnetic Radius: 50918 nT
X-Axis Coverage: 95.4%
Y-Axis Coverage: 100.8%
Z-Axis Coverage: 72.8%

--- Physical Misalignment Analysis ---
```

Sensors are physically misaligned by: 9.86 degrees

Observations: This is a fantastic "golden" calibration. Both the X and Y axes achieved near-perfect ~100% coverage. The Z-axis achieved ~73% on both sensors, which is very typical because fully tumbling a 3-foot wand through the Z-axis (pointing straight up and straight down) is mechanically difficult indoors.

IMU Sensor Readings

The accelerometer and gyroscope readings over the course of the calibration sequence.

Observations:

- The calibration involved a lot of dynamic movement (as expected during a figure-8 or tumbling sweep).
- The accelerometer sees 1g total magnitude properly distributed across the axes depending on orientation.
- The gyroscope properly tracks the rotations. Note: This specific log did not contain a 1-second stationary hold at the end, so the script skipped the FOC (Fast Offset Calibration) for the gyro.

Computed Euler Angles (Sensor Fusion)

Here are the Azimuth and Elevation output from the Madgwick filter during the calibration sequence.

Observations:

- Since the firmware resolved the magnetometer negation bug, we can see the Elevation curve successfully traversing the full mechanical ranges during the tumble without stalling or fighting the IMU!
- The orientation output is incredibly smooth and stable.